EXPLICIT DIOPHANTINE CONSISTENCY EQUATIONS

Reanchored sign constraints and adaptive Pell carriers

A sextic in 34 unrestricted integers,
a degree-30 equation in 13 positive integers,
and a degree-110 equation in 12 positive integers

Research contribution prepared for the ProveIt project

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Abstract

We give three explicit reductions of the current ProveIt Diophantine consistency constructions. First, reanchoring common-sign constraints allows a monic transport elimination while keeping every residual of degree at most three. Seven coordinates are removed from a 41-integer quartic skeleton, yielding a degree-six polynomial in 34 unrestricted integers, compared with the collection's 35-variable sextic. A separately proved complete coefficient adapter reduces the four numeral payloads from 532,479 to 295,388 raw bytes. Second, replacing the main Pell carrier q^5, q^4X by q^2Y, qXY lowers a thirteen-positive-variable equation from degree 34 to degree 30, and from 93 to 92 binary operations. Third, centering both root expressions in an auxiliary norm lowers a twelve-positive-variable equation from degree 118 to degree 110, at the cost of one operation.

The new reductions preserve existence of a finite syntactic ZFC refutation, by composition with explicitly identified source theorems. Thus ZFC proves their unsolvability equivalent to Con(ZFC). The archive supplies literal polynomials, exact coefficient payloads, proofs, independent algebra checks, and provenance. These are local improvements, not a proof of global simplicity or a new end-to-end proof-assistant certificate. A further appendix proves reusable Pell and syntax-encoding lemmas; the proposed direct syntax route remains an incomplete alternative.

Baseline: ProveIt commit
`9cca523d9f27ad39d52993f4065cbd7e58f8bdd4`.
Unknown counts exclude fixed numerals and polynomial abbreviations.

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1 Results and the exact logical claim

The principal equation is the sum of the squares of the nineteen residuals of degree at most three in Section 3. It has exactly 34 unrestricted integer unknowns and exact degree six. The smaller-dimensional companions use positive integers and have higher degrees. These are different useful simplicity criteria; none of these points dominates all the others.

Construction	Domain	Unknowns	Degree	Binary operations
Source 62 sextic	$\mathbb{Z}$	35	6	not compared
New S_{34}	$\mathbb{Z}$	**34**	**6**	not optimized
Source 61 P_{13}	$\mathbb{Z}_{>0}$	13	34	93
Adaptive P_{13}	$\mathbb{Z}_{>0}$	**13**	**30**	**92**
Source 61 product	$\mathbb{Z}_{>0}$	12	118	104
Centered P_{12}	$\mathbb{Z}_{>0}$	**12**	**110**	105
Centered SOS companion	$\mathbb{Z}_{>0}$	12	164	104

Table 1: Comparison with the inspected repository. An operation is a live binary $+, -, \times$ gate, including multiplication by a numeral and each squaring. Register reuse and numeral leaves are free.

The previous twelve-variable SOS companion has degree 172 and 103 operations. Thus the new twelve-variable points exchange one operation for a lower degree; they do not dominate the old points in every metric. The thirteen-variable point improves both degree and operation count.

The baseline already contains nine-positive-variable and ten-unrestricted-integer equations of much larger degree, and a quartic in 41 unrestricted integers. In particular, 34 is not a new fewest-variable bound for arbitrary-degree equations. The wave-9 nine-positive family includes degree 1 494 006 at 487 operations and other tradeoffs [11, 10]. The repository's closed executable nine-natural-variable Lean theorem is also a distinct achievement whose proof status must not be confused with that of these native literal equations [12].

1.1 Consistency and provability

Fix the repository's recursive predicate $\mathrm{Ref}_{\mathrm{ZFC}}(p)$ for a finite derivation of falsity from its specified ZFC axiom predicate, and write

$$\mathrm{Con}(\mathrm{ZFC}) \;:\Longleftrightarrow\; \neg\exists p \in \mathbb{N}\ \mathrm{Ref}_{\mathrm{ZFC}}(p).$$

The claimed arithmetical statements, with all coefficient numerals fixed as specified in Section 6, are

$$\mathrm{ZFC} \vdash (\neg\exists \mathbf{x} \in \mathbb{Z}^{34}\ S_{34}(\mathbf{x}) = 0) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}), \tag{1}$$

$$\mathrm{ZFC} \vdash (\neg\exists \mathbf{x} \in \mathbb{Z}_{>0}^{13}\ P_{13}(\mathbf{x}) = 0) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}), \tag{2}$$

$$\mathrm{ZFC} \vdash (\neg\exists \mathbf{x} \in \mathbb{Z}_{>0}^{12}\ P_{12}(\mathbf{x}) = 0) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}). \tag{3}$$

They follow by proving existence of a zero equivalent to existence of a refutation and negating. No consistency assumption enters a coefficient definition or witness construction.

Here ZFC-provability is justified by ordinary mathematics formalizable in ZFC, composed with the cited source arithmetic and compiler theorems. This contribution proves the new reductions, not the entire upstream compiler again. It supplies neither a new end-to-end Lean/Rocq certificate nor a derivation encoded in the repository's internal ZFC calculus. Finite scripts audit identities and data; they cannot prove consistency by testing candidate solutions.

1.2 The fixed ports

The four ports β, K_0, M_C, M_F are fixed integer numerals, not unknowns. The archive supplies losslessly compressed, minimal unsigned little-endian encodings and deterministic finite reconstruction, with raw-byte hashes.

The directly inherited S_{34} instance uses the entire source-34 tuple with $\beta = 2^{1064921} - 1$. The preferred compact instance, and both positive companions, use the entire compact source-44 tuple with $\beta = 2^{590069} - 1$. A full adapter for the compact integer instance is proved in Section 3.3. In both tuples M_F is already the shifted high mask. One must never mix isolated ports from these different tuples.

2 Optimizing common-sign constraints

For integers s, t, v, define

$$\mathrm{Pos}(s, t, v) = 1 + s^2 + t^2 + v(v + 1).$$

This is always positive. Conversely, every positive integer m has this form. Legendre's three-square theorem applies to $4m - 3$, which is 1 or 5 mod 8. Exactly one square in such a representation is odd modulo four. Writing it as $(2v + 1)^2$ and the other two as $(2s)^2, (2t)^2$ gives the formula after division by four.

Lemma 2.1 (Changing the sign-constraint tree)**.** *Let $x_1, \ldots, x_m$ be integers, $m \geq 2$, and T any spanning tree on their indices. The inequalities*

$$x_i x_j > 0 \qquad (\{i, j\} \in E(T))$$

are equivalent to all the x_i having one common nonzero sign. Replacing T by another spanning tree preserves existential solvability when each edge inequality is represented by its own fresh ternary Pos *witness.*

Proof. A positive edge product gives its endpoints the same nonzero sign, and connectivity propagates that sign to every vertex. The converse is immediate. Surjectivity of Pos supplies the witnesses in both directions. □

Witness triples generally have to be reselected: their represented products change. This is an existential equivalence, not an identity with every old coordinate left fixed.

Proposition 2.2 (Best tree for fixed polynomial degrees)**.** *If the nonzero expressions substituted for x_i have degrees d_i, then*

$$\min_T \max_{\{i,j\} \in E(T)} (d_i + d_j) = \min_i d_i + \max_i d_i.$$

A star centered at a minimum-degree expression attains this bound. The resulting positivity residuals have degree at most $\max(2, \min d_i + \max d_i)$.

Proof. Every tree gives a maximum-degree vertex a neighbor of degree at least $\min d_i$. The proposed star attains this lower bound. The positivity polynomial has degree two. □

After the transport substitution below, F, α, C have degrees $2, 2, 1$. The old edges $F\alpha, FC$ have degrees four and three. The new edges $C\alpha, CF$ both have degree three. Reanchoring the guards therefore permits a sextic sum of squares. This elementary principle is a useful optimization method, with no claim of historical priority.

3 A sextic in 34 unrestricted integers

3.1 The complete polynomial

Supply the following 25 unrestricted integers,

$$J, C, z, f, h, r, v, w, t, \eta, \zeta, y, \gamma, o, j,$$
$$N, E, c, a, D, V, I, B_{\mathrm{a}}, \Delta, u,$$

and three further triples $\mathbf{p}, \mathbf{s}, \mathbf{e} \in \mathbb{Z}^3$. There are $25 + 9 = 34$ unknowns. In the literal files v, w, t are named `v_scale`, `w_scale`, `tau_gap`, and the triples are `CA_pos*`, `FC_pos*`, `eta_zeta_pos*`.

With one complete fixed tuple, put

$$\begin{aligned} q &= \beta J + 1, & k &= \eta + \zeta, & H &= 4a + 3, \\ \widehat{F} &= (K_0 + 4w)C - z(q-1), & \widehat{\alpha} &= q - \widehat{F} - C, & & \\ \widehat{R} &= (q - \widehat{F})(N - q) - C(q^2 - 1) + (M_C + qM_F)J. & & & & \end{aligned} \tag{4}$$

These are abbreviations, not extra coordinates. In particular, N remains an independent unknown; substituting q^3 into $\widehat{R}$ prematurely would raise its degree.

Define nineteen residuals:

$$\begin{aligned} L_1 &= N - q^3, \\ L_2 &= 2u^2 - 4wq, \\ L_3 &= a - E - (2v+1)N, \\ L_4 &= 2u^2(E - a) + E, \\ L_5 &= c - \eta + (E - a)k, \\ L_6 &= t^2 + (2t - k)E(c - \eta) - 1, \\ L_7 &= D - 2u^2 - ac - \gamma H, \\ L_8 &= \Delta - a^2 - 4a - 3, \\ L_9 &= D^2 - \Delta c^2 - 1, \\ L_{10} &= k - 1 - hE - \widehat{R}, \\ L_{11} &= V + c - fo, \\ L_{12} &= V - cj - hE + k - 1, \\ L_{13} &= I - (2r+1)c^2, \\ L_{14} &= B_{\mathrm{a}} - I^2, \\ L_{15} &= B_{\mathrm{a}} - \Delta(f^2 - 1), \\ L_{16} &= B_{\mathrm{a}}(V^2 - y^2) + y^2 - 1, \\ L_{17} &= C\widehat{\alpha} - \mathrm{Pos}(\mathbf{p}), \\ L_{18} &= \widehat{F}C - \mathrm{Pos}(\mathbf{s}), \\ L_{19} &= \eta\zeta - \mathrm{Pos}(\mathbf{e}). \end{aligned} \tag{5}$$

The single Diophantine equation is

$$\boxed{S_{34} = \sum_{\nu=1}^{19} L_\nu^2 = 0, \qquad \text{all 34 inputs in } \mathbb{Z}.} \tag{S34}$$

There are no implicit divisions, variable exponents, or additional root witnesses in this definition.

3.2 Exact local equivalence

Let Q_{41} denote the sum of the 26 squared residuals in the pinned source-64 `integer41/quartic41.json`. Its byte-exact copy is in `integer34/upstream/`; the Git blob is `521f16a310f49e736ea8ffac3fe13847ff71adcb`.

Theorem 3.1. *For any fixed integer values of the four coefficient ports, the two polynomial skeletons satisfy*

$$\exists \mathbf{x} \in \mathbb{Z}^{34}\ S_{34}(\mathbf{x}) = 0 \quad \Longleftrightarrow \quad \exists \mathbf{y} \in \mathbb{Z}^{41}\ Q_{41}(\mathbf{y}) = 0.$$

Proof. Replace the source coordinate α by $C = q - F - \alpha$, whose inverse is $\alpha = q - F - C$. Use Lemma 2.1 to replace $F\alpha > 0, FC > 0$ by $C\alpha > 0, FC > 0$, reselecting the first witness triple.

The transport row becomes monic in F:

$$(K_0 + 4w)C - z(q-1) - F = 0.$$

Eliminate $F = \widehat{F}$. The two new guards $C(q - \widehat{F} - C) > 0, \widehat{F}C > 0$ are cubic; the old first guard would have become quartic.

Six other monic rows eliminate

$$L = q^2 - 1, \quad X = 2u^2, \quad C_2 = c^2, \quad F_2 = f^2, \quad g = y^2 - V^2, \quad P = k + E(c - \eta).$$

Retain N, so the source $N = q(L+1)$ becomes $N = q^3$. Change the first Pell-root coordinate by the polynomial automorphism $\tau = t + E(c - \eta)$. Its source row $\tau^2 - P(P-k) - 1 = 0$ becomes

$$t^2 + (2t - k)E(c - \eta) - 1 = 0;$$

the nominal quartic terms cancel.

All remaining rows are exact substitutions. The packing identity is

$$((q-1)(q-F) + \alpha)(q^2 - 1) = (q - F)(N - q) - C(q^2 - 1)$$

when $\alpha = q - F - C$ and $N = q^3$. This proves that every source zero projects to a new zero.

Conversely, a zero of S_{34} makes every residual zero. Restore $F = \widehat{F}, \alpha = \widehat{\alpha}, \tau = t + E(c - \eta)$ and the six coordinates in (3.2). The guards give $C\alpha > 0, FC > 0$, hence $F\alpha > 0$. Choose a source positivity triple for the latter product. Every eliminated row holds by definition, and all retained rows hold by the stated identities. This gives a Q_{41} zero. □

The complement and gap changes preserve dimension. Seven rows and seven coordinates are eliminated, so $41 - 7 = 34$ and $26 - 7 = 19$. Every remaining variable occurs for each of the two specified coefficient tuples.

Corollary 3.2. *The degree of S_{34} is exactly six for every coefficient specialization. With the original source-34 tuple, (1) follows from the source Q_{41} theorem.*

Proof. Every L_ν has degree at most three. Only L_{13} involves r, and

$$[r^2 c^4] S_{34} = 4.$$

This proves exact degree six. The source theorem $\exists Q_{41} = 0 \iff \exists p\ \mathrm{Ref}_{\mathrm{ZFC}}(p)$ [1, 2], composed with Theorem 3.1, proves the consistency equivalence. □

3.3 A complete compact-tuple adapter

The same skeleton also accepts the *entire* compact source-44 tuple, as a separately identified instance. This is not inferred from small numerical bounds alone. Its semantic adapter and canonical parity promises are explicitly present in source 44 and source 36 [6, 7].

Theorem 3.3. *With the compact tuple defined in Section 6,*

$$\exists \mathbf{x} \in \mathbb{Z}^{34}\ S_{34}(\mathbf{x}) = 0 \quad \Longleftrightarrow \quad \exists p \in \mathbb{N}\ \mathrm{Ref}_{\mathrm{ZFC}}(p).$$

Proof. Restore the Q_{41} skeleton using Theorem 3.1. Write $Q = |q|$, $X = 4qw = 2u^2$, $Y = (2v+1)q^3$, and let the ordinary strong multiplier be $i = 2r+1$. The actual compact coefficients satisfy

$$\beta = 2^d - 1, \quad d = 590069 \geq 3, \quad \beta \geq 17, \quad 0 < M_C < \beta < M_F < 2\beta, \quad K_0 \geq 0.$$

The common-sign simplex therefore gives the same sign-free source-64 bounds

$$Q \geq 16, \qquad Q(Q^2-1) < R < Q^4 - 2, \qquad R > 3Q + 3.$$

The main norm modulo $4a+3$ excludes $X = 0$, so $X \geq 4Q$ and $|XY| \geq 4Q^4 > 3R+1$. If both η, ζ were negative, the first Pell congruence would imply $n > R$, while the ordinary strong and signed auxiliary lemmas imply $p = R$ with $p > n$; this is impossible. Thus $\eta, \zeta > 0$.

The rest of the source-64 signed Pell argument uses only these bounds and the literal norm and quotient rows. It rules out $Y < 0$, recovers the exact two ranks, selects the positive main root, and proves $X = 2^R$. None of these steps depends on the old block width or old native geometry.

Now $q \mid 2^R$. The elementary Mersenne sign lemma $-2^t \not\equiv 1 \bmod (2^d - 1)$, $d \geq 3$, excludes negative q. To prove that lemma, reduce t modulo d: the residue 2^r, $0 \leq r < d$, lies below $2^d - 2$. Thus $q = 2^{t_{\mathrm{rad}}} > 0$, and $q \equiv 1 \bmod (2^d - 1)$ also gives $d \mid t_{\mathrm{rad}}$. Hence $q = B^{N_{\mathrm{tab}}}$ for the compact base $B = 2^d$.

The source half-binomial and inverse-population stages restore the compact native masks and the shifted transport

$$(K_0 + w_{\mathrm{source}})C = F + z(q-1), \qquad w_{\mathrm{source}} = X/q.$$

The rank bound gives $z > 0$; the unshifted convention uses $z_{\mathrm{old}} = z + 1$. The full source-44 native proof, Sections 5–7, now applies to its own D_C, D_R, masks, and rotation geometry. In particular its $D_R = \rho^{42}$ is used, not the source-34 $D_R = \rho$. It decodes a genuine accepting blank K215 tableau. The inherited literal-machine calibration gives a finite ZFC refutation.

For completeness, start with such a refutation. Source 44 constructs a padded tableau with $N_{\mathrm{tab}} = 581197^{\kappa} > 1$ odd and $t_{\mathrm{rad}} = dN_{\mathrm{tab}}$ odd. Its actual-rank controller tunes the rotation, not merely a surrogate rank. Source 44 Section 6 explicitly preserves

$$\mathrm{pc}(R) = 3t_{\mathrm{rad}} + 2, \qquad R \equiv 3 \pmod 4, \qquad R > t_{\mathrm{rad}},$$

and the square- and odd-scale promises of source 36 Section 6. The latter spells out $v_2(Y) = 3t_{\mathrm{rad}}$, hence $s = Y/q^3$ is positive odd, and the canonical ordinary strong multiplier i is positive odd. Since $R - t_{\mathrm{rad}}$ is positive and even, $w_{\mathrm{source}} = 2^{R - t_{\mathrm{rad}}}$ is divisible by four. Choose

$$w = w_{\mathrm{source}}/4, \quad u = 2^{(R-1)/2}, \quad v = (s-1)/2, \quad r = (i-1)/2.$$

The remaining positive canonical coordinates and their exposed quotients satisfy the Q_{41} skeleton. Supply the three ternary positivity triples, then apply Theorem 3.1. □

The complete interface-by-interface proof and its source blobs are recorded in `integer34/COMPACT_S34_ADAPTER.md`. The two literal files remain separate: `sextic34.json` retains its source-34 binding, while `sextic34_compact.json` selects the whole compact tuple. The compact instance uses 295,388 raw coefficient bytes, against 532,479: a reduction of 237,091 bytes, or 44.5259%. The degree certificate is independent of the ports and remains $[r^2c^4] = 4$.

4 An adaptive carrier in thirteen positive unknowns

4.1 Complete polynomial

Use the compact coefficient tuple. Supply exactly

$$j, F, \alpha, f, i, T, X, Y, r, \eta, \zeta, u, \gamma \in \mathbb{Z}_{>0}.$$

Every other symbol is a polynomial abbreviation:

$$\begin{aligned}
J &= 3j + 1, \qquad q = \beta J + 1, \qquad C = q - F - \alpha,\\
E &= XY, \qquad k = \eta + \zeta, \qquad c = kY + \eta,\\
a &= E + Y, \qquad H = 4a + 3, \qquad \Delta = a^2 + H,\\
\ell &= EkY, \qquad b_0 = X + ac, \qquad I = ic^2,\\
R &= ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J,\\
r_0 &= k - R - 1, \qquad U = q^2Y,\\
D_* &= Ub_0 + H(U\gamma + qXY - r_0),\\
W &= (q-1)k + 1, \qquad M_0 = q^3(q-1),\\
B_t &= (qK_0 + X)C - qF,\\
Z &= M_0(\ell + k + r) + (q-1)Y - q^3WB_t,\\
K_{\mathrm{a}} &= \Delta(f^2 - 1), \qquad V = f^2(cT - R) - c.
\end{aligned} \tag{6}$$

The four residuals are

$$\begin{aligned}
A_0 &= Z^2 - M_0^2(\ell(\ell + k) + 1),\\
A_1 &= D_*^2 - U^2(1 + \Delta c^2),\\
A_{\mathrm{a}} &= (V + u)^2 - K_{\mathrm{a}}u(2V + u) - 1,\\
A_{\mathrm{s}} &= f^2 - \Delta I^2 - 1.
\end{aligned} \tag{7}$$

Set

$$\boxed{P_{13} = A_0^2 + A_1^2 + A_{\mathrm{a}}^2 + A_{\mathrm{s}}^2 = 0, \qquad \text{all 13 inputs in } \mathbb{Z}_{>0}.} \tag{P13}$$

The mathematical change from source 61 is precisely

$$\boxed{U = q^5,\ q^4X \quad \longmapsto \quad U = q^2Y,\ qXY}$$

inside the main carrier. Shared registers in the literal graph provide the stated operation count.

4.2 Inherited Pell and compiler interfaces

For $A \geq 2$, define the ordinary Pell sequences by

$$\chi_A(n) + \psi_A(n)\sqrt{A^2 - 1} = (A + \sqrt{A^2 - 1})^n.$$

We use Pell classification, composition, strong divisibility, and recurrence congruences. The less elementary source interface is the signed auxiliary-index recovery theorem [4, 3, 5].

Lemma 4.1 (Source signed auxiliary-index interface)**.** *Let $A \geq 2$, $\Delta = A^2 - 1$, $c = \psi_A(p)$, $f = \chi_A(m)$, with $c \mid m$, $m > 2p$, and $0 < p, R < c/2$. Suppose $M > 1$, $c \mid M$, and $M^2 = \Delta(f^2 - 1)$. An integer pair V, y in the nontrivial branch $|V| > 1$, satisfying*

$$M^2V^2 - (M^2 - 1)y^2 = 1, \qquad V \equiv -c \pmod{f}, \qquad V \equiv -R \pmod{c},$$

forces $p = R$.

The cited proof uses odd-index Chebyshev quotient identities and the plus-congruence step-down theorem. In the application below, every size, sign, and divisibility hypothesis is checked before the conclusion is used. No hypothesis $q \mid X$ belongs to this interface.

The second inherited interface is semantic. The normalized positive kernel recovered below, with the whole compact tuple, has a solution exactly when there is a finite ZFC refutation. A refutation supplies canonical witnesses with

$$\begin{gathered} q = B_{\text{native}}^{N_{\text{tab}}}, \quad B_{\text{native}} = 2^{590069}, \quad J = 1 + B_{\text{native}} + \cdots + B_{\text{native}}^{N_{\text{tab}}-1}, \\ N_{\text{tab}} = 581197^v > 1 \text{ odd}, \quad R \equiv 3 \pmod 4, \quad X = qw = 2^R, \quad Y = sq^3, \\ c = \psi_{a+2}(R), \qquad k = 2\psi_{2XY^2+1}\big((R+1)/2\big). \end{gathered} \tag{8}$$

Padding permits these choices. The same source supplies the strong and squared-modulus auxiliary witnesses used for completeness below [4, 6]. These are mathematical interfaces, not empirical conclusions from finite sampling.

4.3 Soundness in the necessary order

Suppose all four residuals vanish. We repeatedly use the elementary fact that a rational number with integer square is an integer: in lowest terms, its denominator's square divides its numerator's square, forcing the denominator to be one.

First carrier and recovered signs. From $A_0 = 0$, rational-root integrality gives $M_0 \mid Z$ and an integer, possibly signed, $\tau = Z/M_0$, with $\tau^2 = \ell(\ell + k) + 1$. Reducing $M_0 \mid Z$ modulo q^3 and modulo $q - 1$ gives

$$q^3 \mid Y, \qquad q - 1 \mid B_t,$$

since $\gcd(q, q - 1) = 1$ and $W \equiv 1 \bmod (q - 1)$. Put $s = Y/q^3 > 0$, $t = B_t/(q - 1)$. The carrier becomes

$$\tau = \ell + k + r + s - Wt.$$

The norm gives $|\tau| < \ell + k$, so $Wt > 0$, hence $t > 0$ and $B_t > 0$. If $C \leq 0$, however, $(qK_0 + X)C - qF < 0$. Thus $C > 0$, before any use of $q \mid X$. The authentic coefficient bounds imply

$$q \geq 16, \qquad 3q < R < q^4 - 2.$$

Also $\beta \equiv J \equiv 1 \bmod 3$, so

$$q \equiv 2 \pmod 3, \quad H \equiv 3 \pmod q, \quad \gcd(q, H) = 1.$$

Positive main root and the correct submodulus. We have $U = q^2Y \geq q^5$, $r_0 < k$, $c > kY$, and $UaY > H$. Therefore

$$D_* > Uac - Hk > k(UaY - H) > 0.$$

From $A_1 = 0$, obtain $D = D_*/U \in \mathbb{Z}_{>0}$, with

$$D^2 - \Delta c^2 = 1, \qquad U \mid H(qXY - r_0).$$

Now $q^5 \mid U$, while $q^4 \mid qXY$. Using only the q-power submodulus and (4.3) gives

$$q^4 \mid r_0.$$

There is no assertion that $\gcd(U, H) = 1$; cancelling against all of U at this stage would be unjustified.

Main index before the scale relation. Put $A = a + 2$ and $P_0 = 2XY^2 + 1$. The two Pell classifications give

$$|\tau| = \chi_{P_0}(n), \quad k = 2\psi_{P_0}(n), \qquad D = \chi_A(p), \quad c = \psi_A(p), \qquad n, p \geq 1.$$

Since $c > 1$, $p \geq 2$. If $n = 1$, then $k = 2$; positivity forces $\eta = \zeta = 1$ and $c = 2Y + 1$. But $c \geq \psi_A(2) = 2A \geq 4Y + 4$, a contradiction. Thus $n \geq 2$, and

$$c > kY \geq 4P_0Y > 8XY^3 \geq 8q^9 > 2R.$$

Since $P_0 > A$ and $c > k$, monotonicity gives $n < p$. Pell growth also gives $c > 2p$.

The strong equation gives $f = \chi_A(m)$, $I = \psi_A(m)$. As $c^2 \mid I$, strong divisibility gives $p \mid m$. Write $m = pv$. Composition gives $I = c\psi_D(v)$; the congruence $\psi_D(v) \equiv vD^{v-1} \bmod c$ and $\gcd(D, c) = 1$ imply $c \mid v$. Hence $c \mid m$, and $m \geq pc > 2p$.

Set $M = \Delta I$, $y = |V + u|$. Then $M^2 = K_{\mathrm{a}}$, $c \mid M$, and the auxiliary residual is

$$M^2V^2 - (M^2 - 1)y^2 = 1.$$

The exposed congruences are $V \equiv -c \bmod f$ and $V \equiv -R \bmod c$, because $f^2 \equiv 1 \bmod c$. Here $f > 2c > 4$, excluding $V = \pm 1$; the norm excludes $V = 0$. Thus every hypothesis of Lemma 4.1 holds, and $p = R$. No preceding step assumed $q \mid X$.

First index and scale recovery. The first recurrence gives $k \equiv 2n \bmod 4XY^2$. Its modulus is a multiple of q^4, since $q^3 \mid Y$. Together with (4.3), this gives $q^4 \mid 2n - R - 1$. But $1 \leq n < p = R < q^4$, so $|2n - R - 1| < q^4$. Hence $2n = R + 1$.

Consequently $r_0 = k - 2n$ is a positive multiple of $4XY^2$. The adaptive modulus now has the decisive property

$$\frac{4XY^2}{U} = \frac{4XY}{q^2} \in \mathbb{Z}.$$

Thus $U \mid r_0$, and (4.3) implies $U \mid HqXY$, hence $q \mid HX$. Equation (4.3) gives $q \mid X$. The numbers $w = X/q$ and $h = r_0/E$ are positive integers. Also

$$\gamma_{\mathrm{old}} = \gamma + w - r_0/U$$

is integral and $D = X + ac + H\gamma_{\mathrm{old}}$. Since $D > (a + 1)c$ and $c > X$, it is positive.

Restore the transport and source kernel. Now $B_t = q((K_0 + w)C - F) > 0$. Since $q - 1 \mid B_t$, a positive integer z_{new} satisfies $(K_0 + w)C - F = (q - 1)z_{\mathrm{new}}$. With $z_{\mathrm{old}} = z_{\mathrm{new}} + 1$,

$$(K_0 + w)C + (q - F) - z_{\mathrm{old}}(q - 1) = 1.$$

Use $T_{\mathrm{old}} = fT > 0$ and $y = |V + u| > 0$. Then $c(T_{\mathrm{old}}f - 1) - Rf^2 = V$. All source coordinates and relations are restored. The inherited kernel soundness gives a finite ZFC refutation.

4.4 Completeness and a simpler positive gap

Start with canonical compact refutation witnesses (8). Their J is 1 mod 3, giving the positive integer $j = (J-1)/3$. The unchanged first-carrier gap is

$$r = Wt + \tau - \ell - k - s, \qquad t = q(z_{\text{old}} - 1) \geq q.$$

Source growth gives $z_{\text{old}} \geq 2$, $\tau > \ell$, and $k > XY > s$; hence $r > qW - k - s > 0$. Substitution gives $Z = M_0\tau$.

For the canonical strong root and $M = \Delta I$, the squared-modulus source construction, with $R \equiv 3 \bmod 4$, gives

$$V_* = \chi_M(R)/M > 1, \qquad y_* = \psi_M(R) > V_*,$$

and $V_* \equiv -c \bmod f^2$, $V_* \equiv -R \bmod c$. Therefore

$$T = \frac{V_* + c + Rf^2}{cf^2} > 0, \qquad u = y_* - V_* > 0$$

are integers and restore the auxiliary equation.

Only the main carrier changes its witness:

$$\gamma = \gamma_{\text{old}} - w + \frac{r_0}{q^2 Y}.$$

This is integral. Since r_0 is a positive multiple of $4XY^2$, and $X = qw$, $Y = sq^3$,

$$\frac{r_0}{q^2 Y} \geq \frac{4XY}{q^2} = 4wsq^2 > w.$$

Thus $\gamma > \gamma_{\text{old}} > 0$. The predecessor's stronger estimate $\gamma_{\text{old}} > w$ is unnecessary. Now $D_* = UD$, and all four residuals vanish. This proves completeness and (2).

4.5 Exact degree and arithmetic cost

Give all supplied variables degree one and ports degree zero. Put

$$q_0 = 3\beta j, \quad k_0 = \eta + \zeta, \quad R_0 = q_0^3(q_0 - F), \quad B_0 = (K_0 q_0 + X)(q_0 - F - \alpha) - q_0 F.$$

Exact expansion gives

$$[A_0]_{15} = -2q_0^8 k_0^2 XY^2 B_0, \tag{9}$$

$$[A_1]_{13} = 8q_0^2 k_0 X^2 Y^4 (q_0^2 Y\gamma + R_0). \tag{10}$$

The nominal degree-sixteen terms of A_0 cancel. The four residual degrees are $15, 13, 13, 14$. Only A_0^2 reaches degree 30, giving

$$[P_{13}]_{30} = 4q_0^{16} k_0^4 X^2 Y^4 B_0^2, \qquad [j^{16}\eta^4 X^4 Y^4 \alpha^2] P_{13} = 4(3\beta)^{16} > 0.$$

Thus the degree remains exactly 30 with the actual fixed numerals. The literal graph has 92 live gates: 51 multiplications, 24 additions, 17 subtractions.

5 A twice-centered norm in twelve positive unknowns

5.1 The complete polynomial

Supply exactly

$$j, F, \alpha, i, T, X, Y, r, \eta, \zeta, n, \gamma \in \mathbb{Z}_{>0}.$$

Use the same outer definitions and A_0, A_1, but set $J = 6j + 1$. The inputs f, u are absent. Define

$$\begin{aligned} I &= ic^2, & \mathcal{D} &= 1 + \Delta I^2, & M &= \Delta I, & d &= M^2 - 1, \\ b &= cT - R, & V &= \mathcal{D}b - c, & \nu &= n + I(b + a + 2), & & \\ A_* &= M\left[d(\nu^2 + \mathcal{D}) + 1 - V^2\right] - 2d\nu, & & & & & & \\ B_* &= 2d(M\nu - 1), & \mathcal{N} &= A_*^2 - B_*^2\mathcal{D}. & & & & \end{aligned} \tag{11}$$

Here $\mathcal{D}$ is the polynomial square of the missing strong root, not the earlier main Pell root. The single equation is

$$\boxed{P_{12} = (1 + A_0)(1 + A_1)(1 + \mathcal{N}) - 1 = 0, \qquad \text{all 12 inputs in } \mathbb{Z}_{>0}.} \tag{P12}$$

The SOS companion

$$P_{12}^{\mathrm{SOS}} = A_0^2 + A_1^2 + \mathcal{N}^2 = 0 \tag{P12-SOS}$$

has the same positive zero set, with the same $J = 6j + 1$ convention.

5.2 The product is an exact conjunction

Both J, β are odd, so q is even; therefore M_0, U are even. Hence A_0, A_1 are squares modulo four. So is $\mathcal{N}$, since B_* is even. If the three integer factors in (P12) multiply to one, each is ± 1. A factor -1 would require its residual to equal $-2 \equiv 2 \bmod 4$, impossible for a square modulo four. Thus $P_{12} = 0$ exactly when $A_0 = A_1 = \mathcal{N} = 0$.

5.3 Soundness of the centered norm

Positive inputs give $c \geq 3,\ \mathcal{D} > c + 1,\ M > 1,\ d > 0$. The abbreviation ν need not be positive. Nevertheless $M\nu - 1 \neq 0$ for integer ν, so $B_* \neq 0$. At $\mathcal{N} = 0$, let $f = A_*/B_* \in \mathbb{Q}$. Then $f^2 = \mathcal{D} \in \mathbb{Z}$, hence $f \in \mathbb{Z}$. Put $t = \nu - f,\ y = Mt - 1$. Modulo $f^2 = \mathcal{D}$, the exact identities are

$$A_* - B_* f = Md\,t^2 - 2dt + M(1 - V^2), \tag{12}$$

$$M(A_* - B_* f) = d(Mt - 1)^2 - M^2V^2 + 1. \tag{13}$$

Consequently

$$M^2V^2 - (M^2 - 1)y^2 = 1.$$

Since $b \in \mathbb{Z}$ and $V = \mathcal{D}b - c$, either $b \leq 0,\ V \leq -c < -1$, or $b \geq 1,\ V \geq \mathcal{D} - c > 1$. Thus $|V| > 1$, and (5.3) gives

$$d(y^2 - V^2) = V^2 - 1 > 0.$$

Therefore $|y| > |V|$ and $u_{13} = |y| - V > 0$. Take $j_{13} = 2j,\ f_{13} = |f|$, and this u_{13}, retaining all other inputs. The outer J, q, A_0, A_1 agree; the strong and auxiliary equations follow from $f^2 = \mathcal{D}$ and (5.3). Every occurrence of f in (P13) is through f^2. We have produced a positive P_{13} zero.

5.4 Completeness and the positive centered coordinate

Canonical compact J is a repunit of odd length $N_{\mathrm{tab}} > 1$ in the even base $2^{590069} \equiv 2 \bmod 3$. Thus J is odd and $J \equiv 1 \bmod 3$, hence $J \equiv 1 \bmod 6$. The input $j = (J-1)/6$ is therefore a positive integer. This proves that the product finalizer's parity restriction loses no canonical refutation witnesses.

Choose the canonical auxiliary pair

$$V = \chi_M(R)/M > 0, \qquad y = \psi_M(R) > V.$$

Since $R \equiv 3 \bmod 4$, the odd-index recurrence gives $y \equiv -1 \bmod M^2$. For the canonical strong root $f > 0$, set

$$t = (y+1)/M, \qquad \nu = f + t, \qquad n = \nu - I(b + a + 2).$$

These are integral. To prove $n > 0$, put $\lambda = (a+2)I$, $u = y - V > 0$. Using $\mathcal{D} - 1 = MI$ and $V = \mathcal{D}b - c$,

$$n = f - \lambda + \frac{u + b - c + 1}{M}.$$

We have $M > \lambda$, $M > c$, $\mathcal{D} < M^2$, $M \geq 3$. Also $R > 3q$ gives $R \geq 7$. Pell growth $\chi_M(R) \geq M^R$ implies $V \geq M^{R-1} \geq M^6$, whereas

$$\mathcal{D}(M\lambda + c) < M^2(M^2 + M) < 2M^4 < M^6.$$

Thus $b = (V + c)/\mathcal{D} > M\lambda + c$. Equation (5.4) yields $n > f > 0$. Substitution in (12)–(13) gives $A_* = B_* f$, hence $\mathcal{N} = 0$. The outer residuals already vanish, proving completeness and (3).

This chooses fresh canonical auxiliary witnesses. It does not claim that every old witness has $y \equiv -1 \bmod M^2$ or a positive shifted coordinate.

5.5 Exact cancellation and metrics

First treat a, c, i, b, n as independent variables with weights $(2, 2, 1, 4, 1)$, with $\Delta = a^2 + 4a + 3$. Complete sparse expansion gives 875 monomials in $\mathcal{N}$, weighted degree 82, and highest form

$$[\mathcal{N}]_{82} = 4a^{12}c^{20}i^{10}b^2.$$

The nominal degree-86 terms in A_*^2 and $B_*^2\mathcal{D}$ cancel exactly. After composition, put $q_0 = 6\beta j$ and retain k_0, R_0, B_0 as before. Then

$$[\mathcal{N}]_{82} = 4i^{10}k_0^{20}X^{12}Y^{32}R_0^2$$

and

$$[P_{12}]_{110} = -64q_0^{10}i^{10}k_0^{23}X^{15}Y^{38}B_0(q_0^2Y\gamma + R_0)R_0^2.$$

The coefficient

$$[j^{18}i^{10}\eta^{23}X^{16}Y^{39}\gamma\alpha F^2]P_{12} = 64(6\beta)^{18} > 0$$

certifies exact degree 110. The graph has 105 live gates: 58 multiplications, 29 additions, 18 subtractions. The SOS companion has exact degree 164 and 104 gates.

The symmetric auxiliary divisor $M = \Delta ic^2$, combined with the shift $I(b + a + 2)$, cancels the polynomial parts of both root expressions. Changing the divisor split alone does not produce the stated improvement.

6 Exact coefficient numerals and provenance

6.1 Complete compact recipe

The preferred instance uses

$$\rho = 2^{67}, \quad L = 8807, \quad d = 67L = 590069, \quad B = 2^d.$$

Take the ordered 943 tiles (w_i, n_i, e_i, s_i), $0 \le i < 943$, from the included literal `compact_coefficients/native/wang943_tiles.json`. The order matters: tile zero is Start and tile one is Forbidden. The canonical serialized tile-list SHA256 is

```
66b55f1d11fca9fefc93b12fcfc8b4cabb75b08e9fd03311f9e894a1c5e5e81f
```

The builder reconstructs this finite list from all 430 literal machine actions, without executing the machine; the independent auditor begins from the list instead.

Define a 41-bit payload for each tile,

$$v_i = w_i + 2^8 n_i + 2^{17} e_i + 2^{25} s_i + 127 \cdot 2^{34}, \qquad c_i = 1 + 2^{10}(2^{41} - v_i).$$

Let $\mathcal{E}$ contain the selector positions $0, \dots, 942$, the copy and anchor positions in the next table, and the two dummy positions $1063, 1064$.

Native position	Payload bit j	Range
$1023 + h$	h	$0 \le h < 8$
1031	34	west occupancy
$1013 + h$	$8 + h$	$0 \le h < 9$
1022	35	north occupancy
$1003 + h$	$17 + h$	$0 \le h < 8$
1011	36	east occupancy
$983 + h$	$25 + h$	$0 \le h < 9$
992	37	south occupancy
$1095, 2195, 3296$	$38, 39, 40$	three anchors

Table 2: The compact tuple's finite copy-position map.

At a copy or anchor position carrying bit j, set $c_e = 2^{10+j}$; set $c_{1063} = c_{1064} = 0$. There are 986 positions in $\mathcal{E}$. Put $\mu = 2^{51} - 2$ and define the following fixed integers:

$$\begin{aligned}
D_C &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \sum_{e \in \mathcal{E}} c_e(\rho^{4393-e} + \rho^{5495-e}),\\
D_R &= \rho^{42},\\
Z_{\text{pad}} &= (2^{65} - 1) \sum_{e=8792}^{8804} \rho^e + (2^{19} - 1)\rho^{8805},\\
M_C &= B - 1 - \sum_{\substack{e \in \mathcal{E}\\ e \ne 1}} \rho^e - 2\rho^{1063},\\
M_{F,\text{native}} &= 4 + \mu(\rho^{4393} + \rho^{5495}) + \sum_{h=0}^{9} \rho^{1013+h} + \sum_{h=0}^{8} \rho^{1065+h}\\
&\qquad + \rho^{2195} + \rho^{3296} + Z_{\text{pad}}. \qquad (14)
\end{aligned}$$

The polynomial's four fixed ports are exactly

$$\boxed{\beta = B - 1, \qquad K_0 = D_C + BD_R, \qquad M_C = M_C, \qquad M_F = M_{F,\text{native}} + B - 1.}$$

Polynomial leaf	Payload name	Bits	Raw bytes	Gzip bytes
β	`beta.bin.gz`	590069	73759	108
K_0	`K.bin.gz`	592884	74111	8940
M_C	`MC.bin.gz`	590069	73759	193
M_F	`MF_source.bin.gz`	590070	73759	204
Total			295388	9445

Table 3: The preferred complete compact tuple. Compression is a delivery format, not a change in mathematical coefficient height.

Thus (14), the finite tile list, and (6.1) specify actual integers, not unspecified program indices. All sums and exponents in this coefficient recipe are fixed and finite.

Each payload below `compact_coefficients/native/coefficients/` is gzip of the minimal unsigned little-endian integer bytes. It is read literally by

```
value = int.from_bytes(gzip.decompress(path.read_bytes()), "little")
```

The raw hashes, in β, K_0, M_C, M_F order, are

```
0664a868cbfef9869ea05c40a1c445563f685ff857092594c1c1223bc4b21f9d
c6d523bd32af8dd0345db843cdadf6932bfcad186912a6d112314b60657c554e
6007cabd5f5228618e8fb613f386d8fc3692a24520d6e4cd72af0262aae1ade4
48333b0d30a4edeb8579f44286cb59806c5622cb42b367f5b5f9ee827711c737
```

The full upstream binding is `compact_coefficients/native/COEFFICIENT_BINDING.json`, and `POSITIVE_PORTS.json` records the polynomial aliases. Its instance identifier is

```
wang943-complement67-shared-anchor-length8807-prime581197
```

The selector, mask, carry, geometry, and actual-rank controller proofs are in the preserved source-44 `native/PROOF.md`. The controller prime is 581197, with step 581196. No coefficient is computed by waiting for a halting computation.

6.2 The original comparison instance

For the directly inherited S_{34} binding, $\rho = 2^{121}$, $L = 8801$, $d = 1064921$, $B = 2^d$. Its full finite recipe is preserved in `integer_coefficients/README.md`, alongside the exact source constructor, independent decoder, and four raw and compressed payloads. Their total raw size is 532,479 bytes; the largest port has 1,065,043 bits. The complete tuple digest is

```
b61ee51bdd2d9e4274db603b64cb1d092a8f5feeb83f510433876f21fea2c85f
```

The manifest specifies its length-delimited digest convention. This instance remains useful for the direct Q_{41}-to-S_{34} comparison. The compact instance is preferred when the four coefficient sizes are part of the simplicity objective.

6.3 Pinned evidence

All retrievals are pinned to commit `9cca523d9f27ad39d52993f4065cbd7e58f8bdd4`. Historical source pins inside copied files are retained unchanged; provenance ledgers separately record current retrieval paths, SHA256, Git blob SHA1, and byte counts. The compact bundle authenticates 44 copied dependencies; the original tuple bundle authenticates 35. The preserved source proofs include native soundness, canonical completeness, signed Pell recovery, and literal

machine/refutation calibration. Some inherited relative links assume the original repository layout; the ledgers identify their exact upstream targets.

7 Verification and repository integration

The archive separates finite algebra checks from unbounded mathematical proofs. The new reductions above establish existence equivalence using explicit source theorems. The independent standard-library scripts establish:

1. All 26 source-row restoration identities for S_{34}, including the seven eliminated rows; inverse complement and gap identities; 34 live inputs; nineteen residuals of degree at most three; and the exact coefficient $[r^2c^4] = 4$.
2. A complete expansion of S_{34} into 437 monomials when the four coefficient ports remain independent symbols. This is not a monomial-minimality assertion after specialization.
3. Exact identities for the positive carriers, graph liveness, gate ledgers, highest-degree forms, and both finalizers. The checker reads the literal graphs and imports no builder.
4. Complete expansion of the 875-term centered norm in an independent sparse polynomial implementation, including the degree-86 cancellation and the two root-recovery identities.
5. Reconstruction of all actual coefficient numerals from the pinned finite data, with raw-byte hashes and the original gzip hashes. No small surrogate coefficients are used.
6. Literal equality of the original and compact sextic polynomial skeletons, with separately authenticated complete port bindings and adapter proof identities.

The compact native auditor checks all 430 machine actions, 66,062 basis identities, 38,663 single-bit mutations, and 87,380 finite complement models. The original tuple audit checks 69,230 basis identities and all 8,801 digits of its transport constant. These finite diagnostics supplement, and do not replace, the universal native proofs.

From the extracted package root, run

```
python3 verify_all.py
python3 -O verify_all.py
latexmk -pdf -interaction=nonstopmode -halt-on-error report.tex
```

Python 3.10 or later and ordinary TeX Live packages suffice. The finite checkers run normally and with optimization; essential validation does not depend on removable `assert` statements. The top-level receipt records individual process outcomes and logs. Exact sparse expansions and checked result JSON files are included.

The new local reductions are suitable for additive integration as a new proposal directory; the archive changes no existing repository file. Its `README.md` identifies the preferred equation, directory layout, exact commands, comparison basis, and proof dependencies. Licenses and upstream notices are preserved. A manifest hashes the delivered files.

7.1 A concrete formalization target

Theorem 3.1 can be formalized independently of the enormous numerals and computation encoding: prove the sign-tree lemma, the positivity representation, and the polynomial substitutions over symbolic integer ports. That local theorem can later be composed with a formal source equation. The carrier proof similarly identifies precise divisibility and positivity lemmas.

Neither this package nor its finite verification closes the native compiler semantic bridge in a proof assistant, recertifies the machine's full formal calibration, or supplies an internally encoded ZFC derivation. The repository's existing distinctions between those targets remain in force [12].

8 Search decisions and remaining work

The request called for a simpler equation and a serious attempt at a new approach. The constructions here improve the inspected frontier in several concrete metrics. They do not establish global simplicity, literature-wide priority, or a new fewest-variable record.

Several tempting shortcuts fail at specific obligations:

- Removing the $X = 2u^2$ guard leaves a negative-power branch not justified by the existing computation encoding. The new sextic keeps that guard.
- The adaptive carrier cannot cancel H against all of $U = q^2Y$. Its proof first uses the q^4 submodulus, then the exact Pell index, and only then recovers $q \mid X$.
- Weakening $q^3 \mid Y$ to $q^2 \mid Y$ is not automatically compatible with the native population threshold. A Pell-jet alternative supplies a useful local direction, but gives a larger degree and does not restore that interface.
- Replacing the transport multiplier $(q-1)k+1$ by k only proves $(q-1) \mid kB_t$; the necessary cancellation is unavailable.
- A direct syntax compiler can omit some well-formedness tests by interpreting unused codes conservatively, as proved in the appendix. It still needs an explicit axiom theory, a bidirectional consistency translation, and global unbounded-proof packing.

The most promising further algebraic task is to optimize the sign-constraint tree and retained registers jointly. The N-versus-L choice and the guard reanchoring interact: optimizing either alone misses the sextic reduction. A more ambitious task is a native compiler whose population threshold can use the weaker $q^2 \mid Y$ scale without first restoring the old kernel. Neither task is declared solved here.

A Two reusable arithmetic lemmas

This appendix proves two independent tools. The first permits a direct equational proof compiler to use a simple injective polynomial numbering without checking that every term code is in the numbering's image. The second recovers an exact power of two from a pair of Pell ranks under weaker hypotheses than a radix-dependent formulation would suggest. Neither result, by itself, constructs a new equation for Con(ZFC). No claim of historical priority is made.

A.1 Injective syntax and conservative extra constants

Throughout this subsection, $\mathbb{N} = \{0, 1, 2, \ldots\}$. Fix the first-order equational signature Σ consisting of constants ϵ, ι, binary function symbols U, C, and unary function symbols N, V. A fixed theory $\mathcal{E}$ is a set of equations in finitely many variables; its axioms are understood through substitution instances. The argument applies to any fixed such set, and in particular to a finite set.

We use the ordinary equational calculus: substitution instances of the axioms, reflexivity, symmetry, transitivity, and congruence for each function symbol. A proof of a closed equation can be taken to consist entirely of ground equations. Indeed, replace every variable occurring in a given finite proof by the ground constant ϵ. This preserves the closed conclusion, sends axiom instances to axiom instances, and preserves every inference. This observation concerns a specified syntactic calculus and requires no semantic completeness theorem.

Define integer polynomials

$$J(a,b) = (a+b)^2 + a + 1, \tag{15}$$

$$U(a,b) = 4J(a,b), \qquad C(a,b) = 4J(a,b) + 1, \tag{16}$$

$$N(a) = 4a + 2, \qquad V(a) = 4a + 3. \tag{17}$$

The code of ϵ is 0, and the code of ι is 1. The code $\langle t \rangle$ of a ground Σ-term is obtained by applying the corresponding polynomial at every constructor.

Lemma A.1 (Disjoint constructors and decreasing decoding)**.** The pairing $J : \mathbb{N}^2 \to \mathbb{N}$ is injective and satisfies $J(a,b) > \max(a,b)$. The four constructor images are pairwise disjoint and disjoint from $\{0,1\}$. Every argument of a constructor has a strictly smaller code than its value.

Proof. Put $s = a + b$. For a fixed s, the value of J lies in $[s^2+1, s^2+s+1]$. These intervals are disjoint because $(s+1)^2 + 1 > s^2 + s + 1$. The interval identifies s, after which $a = J - s^2 - 1$ and $b = s - a$ identify the arguments. Also $J(a,b) \geq s^2 + 1 > s$ for $s \geq 1$, and the case $s = 0$ is immediate. The four constructors have distinct residues modulo 4; the two binary constructors have values at least 4 and 5, respectively. The remaining assertions follow directly from their formulas. □

Let $\mathcal{J}$ be the set of natural numbers outside the four constructor images and $\{0,1\}$. Enlarge Σ by a distinct constant κ_m for each $m \in \mathcal{J}$, obtaining Σ^+. Strong recursion defines a decoding map $\delta : \mathbb{N} \to \mathrm{Term}(\Sigma^+)$: decode a constructor value using its unique arguments, decode $0, 1$ as ϵ, ι, and decode $m \in \mathcal{J}$ as κ_m. The preceding lemma proves that this definition terminates. It also proves, for example,

$$\delta(U(a,b)) = \mathsf{U}(\delta(a), \delta(b)), \qquad \delta(N(a)) = \mathsf{N}(\delta(a)),$$

with analogous identities for the other constructors. Induction gives $\delta(\langle t \rangle) = t$ for every original ground term t.

For an axiom term $t(x_1, \ldots, x_h)$, let $\widehat{t}(n_1, \ldots, n_h)$ be its integer-polynomial evaluation under these arithmetic constructors. Arbitrary natural values n_i are allowed. Constructor compatibility gives

$$\delta\big(\widehat{t}(\boldsymbol{n})\big) = t\big(\delta(n_1), \ldots, \delta(n_h)\big).$$

Thus no test that n_i belongs to the image of the original term numbering is needed.

Theorem A.2 (Conservative interpretation of unused codes)**.** Let $s = t$ be a closed Σ-equation. There exists a finite arithmetic equational proof of $(\langle s \rangle, \langle t \rangle)$, using unrestricted natural term codes, polynomial evaluations of $\mathcal{E}$-instances, and the encoded equational rules, if and only if $\mathcal{E} \vdash s = t$ in the specified ordinary equational calculus.

Proof. An ordinary ground proof can be encoded line by line, proving the reverse implication. Conversely, apply δ to the two term codes on every line of an arithmetic proof. The constructor identities show that it becomes an ordinary ground proof in the signature Σ^+. Every arithmetic axiom instance becomes an actual $\mathcal{E}$-instance with ground Σ^+-terms as substitutions.

Now apply the term homomorphism $\rho : \mathrm{Term}(\Sigma^+) \to \mathrm{Term}(\Sigma)$ which fixes every original symbol and sends every κ_m to ϵ. It sends an axiom instance to an axiom instance and preserves each equational rule. It fixes the closed terms s, t. Consequently the transformed finite proof is an original-signature proof of $s = t$. This is a syntactic retraction argument; it assumes neither consistency nor arithmetic soundness of any theory. □

One explicit polynomial for local inference

Represent a conclusion by its pair of term codes (a, b) and two premises by (c, d) and (e, f). For one additional natural number z, put

$$R_0 = (a-b)^2, \tag{18}$$

$$T_0 = (c-e)^2 + (a-d)^2 + (b-f)^2, \tag{19}$$

$$B_0 = (a - U(c,e) - z)^2 + (b - U(d,f) - z)^2, \tag{20}$$

$$W_0 = (a - 4c - 2 - z)^2 + (b - 4d - 2 - z)^2, \tag{21}$$

$$I(a,b,c,d,e,f,z) = z^2(z-1)^2 + R_0T_0B_0W_0. \tag{22}$$

Here U in the two displayed occurrences is the already defined integer polynomial; these are abbreviations, not extra unknowns.

Proposition A.3 (Exact local rule check)**.** For natural term codes a, b, c, d, e, f, the equation $I = 0$ holds precisely when $z \in \{0, 1\}$ and at least one of the following holds:

1. $a = b$, the reflexivity rule;
2. $c = e$, $a = d$, $b = f$, the rule taking $c = d$ and $c = f$ to $d = f$;
3. $(a, b) = (U(c, e), U(d, f))$ if $z = 0$, or $(a, b) = (C(c, e), C(d, f))$ if $z = 1$;
4. $(a, b) = (N(c), N(d))$ if $z = 0$, or $(a, b) = (V(c), V(d))$ if $z = 1$.

The total degree of I is exactly 10. The variable z is its only local auxiliary coordinate beyond the six equation ports.

Proof. Every displayed factor and summand is nonnegative. Thus $I = 0$ forces both $z^2(z-1)^2 = 0$ and $R_0T_0B_0W_0 = 0$. The former is exactly the conjunction $z \in \{0, 1\}$; the latter is exactly the disjunction of the four rule checks. Each sum of squares vanishes precisely when all its summands vanish. This also proves the converse.

The four factors have degrees $2, 2, 4, 2$, so $\deg I \le 10$. Specializing gives the polynomial identity

$$I(t, 0, t, 0, 0, 0, 0) = 2t^4\big((4t^2 + 3t + 4)^2 + 16\big)\big((3t + 2)^2 + 4\big).$$

Its leading coefficient is 288, proving $\deg I = 10$. □

The second rule and reflexivity derive symmetry: use the premises $c = d$ and $c = c$. They then derive ordinary transitivity by first reversing $a = b$ and using $b = a$, $b = c$. Binary congruence of both arguments includes one-argument congruence by using a reflexive second premise. Hence these local rules generate the ordinary ground equational calculus when supplemented with the axiom instances.

To make the finite-proof convention exact, seed the list by the valid reflexive equation $(0, 0)$. Each later line is either an axiom instance or has two earlier premise indices and a natural z satisfying (22). A one-premise rule may use the seed as its unused second premise. This convention excludes circular justifications, including a purported first inference from itself. The premise-index restrictions and the unbounded list are part of the finite-certificate definition; they have *not* been eliminated into the local polynomial I.

What remains to obtain a consistency equation. One must supply an explicit finite equational theory, prove the required bidirectional syntactic consistency translation, materialize its actual axiom terms, and compile the entire unbounded finite-proof certificate into a fixed Diophantine equation. No variable count, degree bound, or arithmetic-operation improvement for that unconstructed endpoint follows from the local degree-10 calculation.

A.2 Exact Pell power recovery without an early radix bound

For an integer $b \geq 2$, define the Pell sequences by

$$\chi_b(0) = 1, \quad \psi_b(0) = 0, \qquad \begin{pmatrix} \chi_b(m+1) \\ \psi_b(m+1) \end{pmatrix} = \begin{pmatrix} b & b^2 - 1 \\ 1 & b \end{pmatrix} \begin{pmatrix} \chi_b(m) \\ \psi_b(m) \end{pmatrix}.$$

In particular, $\psi_b(1) = 1$ and $\psi_b(m+2) = 2b\psi_b(m+1) - \psi_b(m)$.

Lemma A.4 (Recurrence bounds)**.** For every $b \geq 2$ and integer $m \geq 1$,

$$(2b-1)^{m-1} \leq \psi_b(m) \leq (2b)^{m-1}.$$

Proof. The sequence $\psi_b(m)$ is nonnegative and strictly increasing from $m = 0$ onward. This follows inductively from $\psi_b(0) = 0$, $\psi_b(1) = 1$ and the second-order recurrence. For $m \geq 1$, the same recurrence therefore gives

$$(2b-1)\psi_b(m) \leq \psi_b(m+1) \leq 2b\psi_b(m).$$

Iterating from $\psi_b(1) = 1$ proves both bounds. □

Lemma A.5 (Exact quotient identity)**.** Let $a \geq 1$, $A = a + 2$, and $H = 4a + 3$. For every $m \geq 0$,

$$\chi_A(m) - a\psi_A(m) = 2^m + H \sum_{j=0}^{m-1} 2^{m-1-j}\psi_A(j),$$

where the empty sum is zero. In particular, $\chi_A(m) - a\psi_A(m) \equiv 2^m \pmod{H}$.

Proof. Put $W_m = \chi_A(m) - a\psi_A(m)$ and note that $A^2 - 1 = a^2 + H$. The defining recurrences give

$$W_{m+1} = (A-a)\chi_A(m) + (A^2 - 1 - aA)\psi_A(m) = 2W_m + H\psi_A(m).$$

Since $W_0 = 1$, iterating this identity yields the asserted formula. The sign is that of the positive sequence χ_A; it is not a statement obtained by replacing a possibly signed Pell coordinate by its absolute value without further justification. □

Theorem A.6 (Power recovery)**.** Let X, Y be positive integers and $n \geq 2$. Put

$$R = 2n - 1, \quad a = (X+1)Y, \quad A = a + 2, \quad P = 2XY^2 + 1, \quad H = 4a + 3.$$

Suppose

$$c = \psi_A(R), \qquad k = 2\psi_P(n), \qquad kY < c < k(Y+1),$$

and suppose the *positive* main root $D = \chi_A(R)$ satisfies

$$D - ac \equiv X \pmod{H}.$$

Then $X = 2^R$. No hypothesis involving a radix q, divisibility of X or Y by q, or inequalities $X \geq q$, $Y \geq q^3$ is required.

Proof. Both A and P are at least 2. Direct calculation gives

$$(2A-1)^2 - 8P = 4(X-1)^2Y^2 + 12(X+1)Y + 1 > 0.$$

Since $R - 1 = 2(n-1)$, the recurrence bounds imply

$$\begin{aligned}\frac{c}{k} &\geq \frac{(2A-1)^{2(n-1)}}{2(2P)^{n-1}} \\ &= \frac{1}{2}\left(\frac{(2A-1)^2}{2P}\right)^{n-1} > \frac{4^{n-1}}{2} = 2^{R-2}.\end{aligned}$$

The strict ratio bound $c/k < Y + 1$ therefore gives $Y + 1 > 2^{R-2}$. Since $R \geq 3$ and Y is an integer, $Y \geq 2^{R-2}$. Consequently

$$H = 4(X+1)Y + 3 \geq 8Y + 3 \geq 2^{R+1} + 3 > 2^R.$$

Also $0 < X < H$ follows directly from $X, Y \geq 1$.

Lemma A.5 gives $D - ac \equiv 2^R \pmod{H}$. The hypothesis gives the same residue as X. Both X and 2^R lie strictly between 0 and H, so they are equal. □

The lower ratio bound $kY < c$ is natural in the intended applications, although this particular proof uses only the upper bound. If the source instead supplies $D = X + ac + H\gamma$, then after power recovery the exact quotient formula identifies

$$\gamma = \sum_{j=0}^{R-1} 2^{R-1-j}\psi_A(j) > 0,$$

because $R \geq 3$. Thus the same elementary identity also verifies the sign of the recovered quotient. The hypotheses giving the ranks $R = 2n - 1$, the positive main root, and the strict ratio remain essential inputs to this lemma; they are not established by the lemma itself.

Verification boundary. The accompanying standard-library script checks constructor decoding and retraction, local-rule examples and negative controls, the exact degree-10 specialization, recurrence bounds, and the exact Pell quotient identity. Those finite checks are reproducible corroboration. The unrestricted mathematical statements are established by the proofs above; no finite check establishes a consistency equivalence.

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[2] ProveIt, source 64, *A quartic in 44 unrestricted integer variables*, `integer41/PROOF44.md`, same pinned commit. Contains the absolute-radix sign argument and the original source-34 coefficient binding.

[3] ProveIt, source 61, *Thirteen positive unknowns, exact degree 34, 93 operations*, `positive/PROOF13.md`. https://github.com/VladimirReshetnikov/ProveIt/tree/9cca523d9f27ad39d52993f4065cbd7e58f8bdd4/Computability/DiophantineConsistency/con-zfc-diophantine-forms/61-congruence-carriers

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[9] ProveIt, source 62, `quartic42/companions`, degree-six equation in 35 unrestricted integers. https://github.com/VladimirReshetnikov/ProveIt/tree/9cca523d9f27ad39d52993f4065cbd7e58f8bdd4/Computability/DiophantineConsistency/con-zfc-diophantine-forms/62-sign-and-norm

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[12] ProveIt, `Computability/DiophantineConsistency/README.md` and `STATUS.md`. Distinguishes closed executable Lean theorems, native literal equations, and the internal ZFC target. https://github.com/VladimirReshetnikov/ProveIt/tree/9cca523d9f27ad39d52993f4065cbd7e58f8bdd4/Computability/DiophantineConsistency